

District 7 SPSS Comparison 2023 - 2025 Sales

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Model I

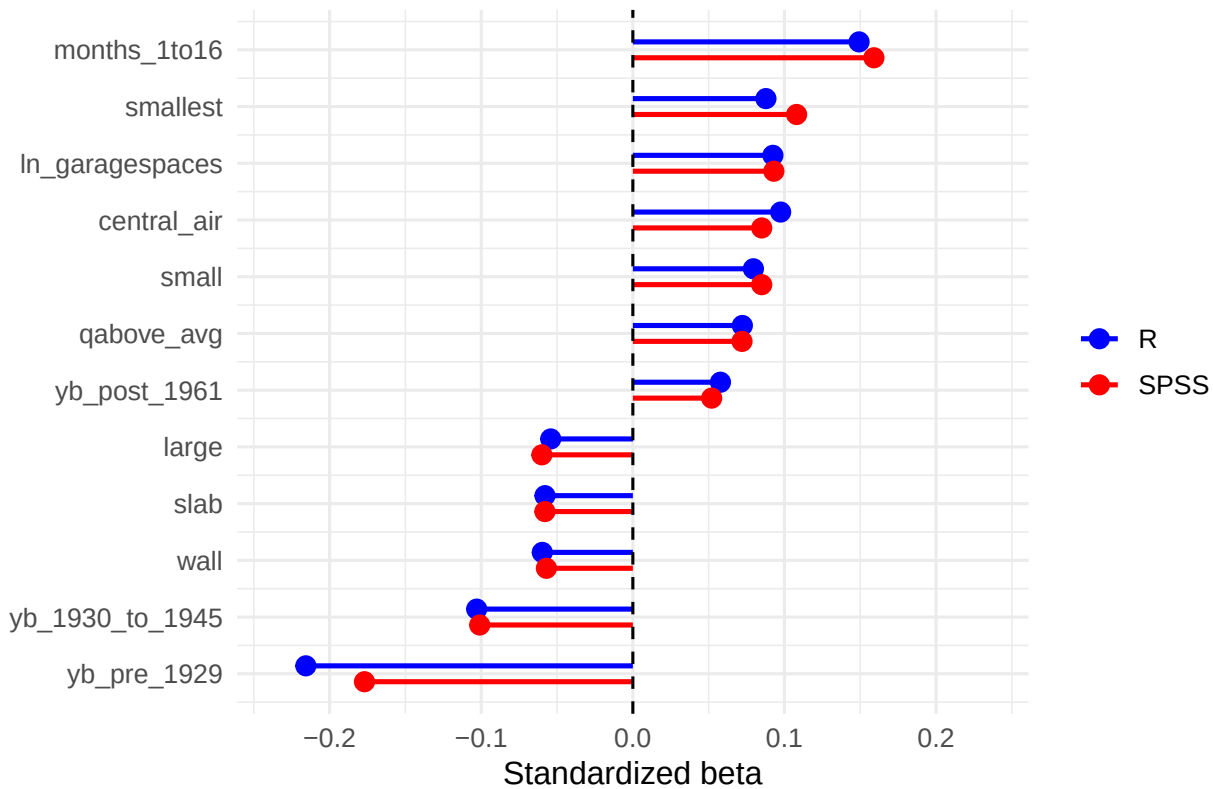
Table 1: District 7 | LN_SPPSF Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1385	1372	13	0.4507	0.1380	0.1300	16.9220
R	1385	1370	15	0.4521	0.1341	0.1247	14.1499
% diff (R vs SPSS)	0.00%	-0.15%	15.38%	0.30%	-2.83%	-4.08%	-16.38%

Table 2: District 7 | Model I Coefficient Comparison: SPSS vs R

	b		Std. Error		Beta		t		Sig		
term	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS	
(constant)	4.107	4.078	0.043	0.038	NA	NA	95.802	106.813	0.000	0.001	
months_1to16	0.014	0.014	0.002	0.002	0.149	0.159	5.746	6.307	0.000	0.001	
central_air	0.120	0.104	0.031	0.031	0.097	0.085	3.797	3.325	0.000	0.001	
ln_garagespaces	0.140	0.142	0.039	0.039	0.092	0.093	3.627	3.674	0.000	0.001	
smallest	0.135	0.167	0.045	0.044	0.088	0.108	3.039	3.762	0.002	0.001	
small	0.079	0.084	0.031	0.029	0.080	0.085	2.572	2.842	0.010	0.005	
qabove_avg	0.377	0.375	0.137	0.134	0.072	0.072	2.758	2.790	0.006	0.005	
yb_post_1961	0.223	0.200	0.099	0.099	0.058	0.052	2.254	2.024	0.024	0.043	
hot_water	0.121	NA	0.065	NA	0.055	NA	1.860	NA	0.063	NA	
large	-0.073	-0.082	0.041	0.039	-0.054	-0.060	-1.772	-2.071	0.077	0.039	
single_family2story	-0.085	NA	0.046	NA	-0.057	NA	-1.853	NA	0.064	NA	
slab	-0.315	-0.315	0.139	0.138	-0.058	-0.058	-2.273	-2.287	0.023	0.022	
wall	-0.481	-0.457	0.203	0.203	-0.060	-0.057	-2.366	-2.251	0.018	0.025	
qbelow_avg	-0.080	NA	0.030	NA	-0.081	NA	-2.697	NA	0.007	NA	
yb_1930_to_1945	-0.106	-0.104	0.028	0.028	-0.103	-0.101	-3.805	-3.777	0.000	0.001	
yb_pre_1929	-0.318	-0.261	0.048	0.043	-0.216	-0.177	-6.623	-6.047	0.000	0.001	
ext_other	NA	-0.123	NA	0.027	NA	-0.120	NA	-4.471	NA	0.001	

District 7 | Model I Beta Coefficient Comparison: SPSS vs R



Model II

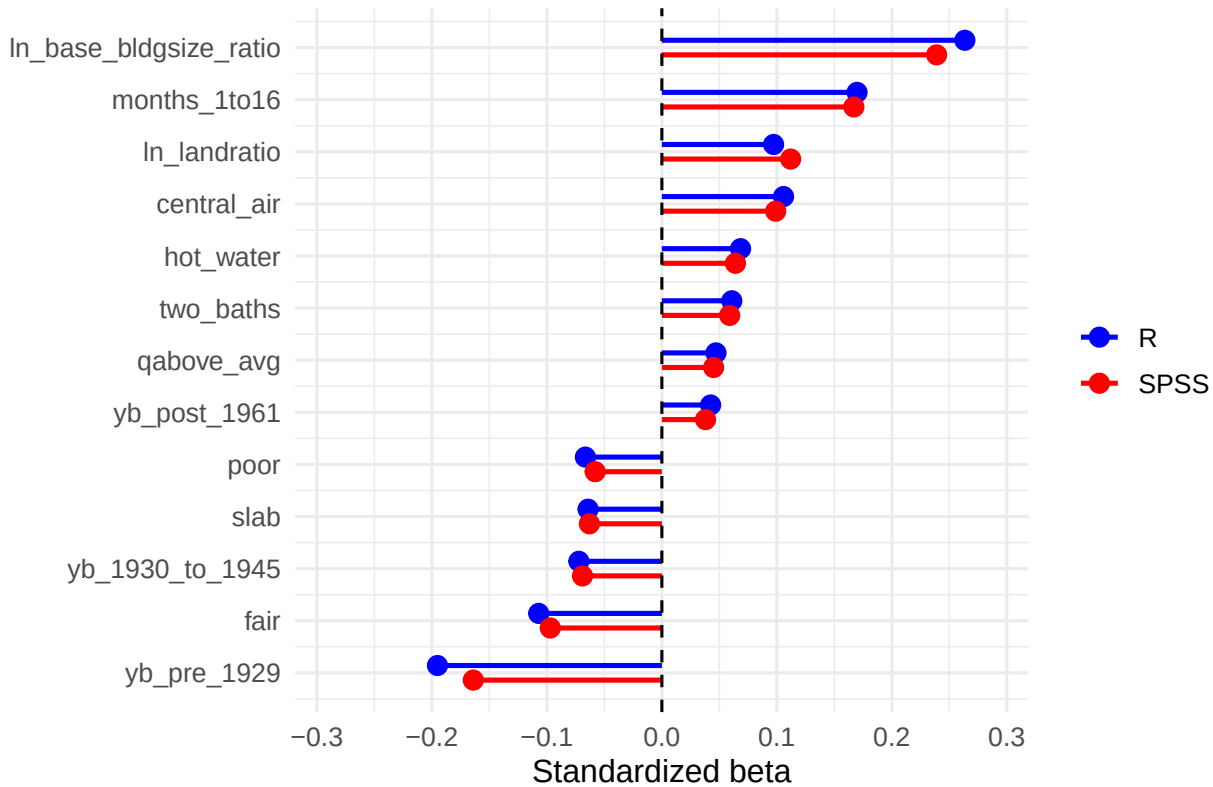
Table 3: District 7 | LN_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1805	1790	15	0.4652	0.2250	0.2190	34.6630
R	1805	1790	15	0.4680	0.2159	0.2094	32.8677
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.59%	-4.04%	-4.38%	-5.18%

Table 4: District 7 | Model II Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	10.998	11.018	0.036	0.031	NA	NA	305.919	356.264	0.000	0.001
ln_base_bldgsize_ratio	0.577	0.524	0.064	0.061	0.264	0.239	9.004	8.625	0.000	0.001
months_1to16	0.017	0.017	0.002	0.002	0.170	0.167	8.083	8.017	0.000	0.001
central_air	0.145	0.135	0.029	0.029	0.106	0.099	4.967	4.648	0.000	0.001
ln_landratio	0.289	0.332	0.067	0.066	0.097	0.112	4.325	5.062	0.000	0.001
hot_water	0.162	0.151	0.060	0.060	0.068	0.064	2.711	2.538	0.007	0.011
two_baths	0.226	0.218	0.095	0.094	0.061	0.059	2.376	2.307	0.018	0.021
qabove_avg	0.282	0.273	0.138	0.137	0.047	0.045	2.049	1.992	0.041	0.047
yb_post_1961	0.178	0.158	0.089	0.089	0.042	0.038	1.990	1.779	0.047	0.075
ln_garagespaces	0.062	NA	0.036	NA	0.037	NA	1.696	NA	0.090	NA
qbelow_avg	-0.052	NA	0.027	NA	-0.049	NA	-1.937	NA	0.053	NA
slab	-0.331	-0.326	0.110	0.109	-0.064	-0.063	-3.010	-2.985	0.003	0.003
poor	-0.209	-0.184	0.067	0.066	-0.067	-0.058	-3.132	-2.772	0.002	0.006
yb_1930_to_1945	-0.081	-0.078	0.026	0.025	-0.072	-0.069	-3.164	-3.079	0.002	0.002
fair	-0.145	-0.132	0.029	0.029	-0.107	-0.097	-5.041	-4.588	0.000	0.001
yb_pre_1929	-0.294	-0.248	0.040	0.040	-0.195	-0.164	-7.335	-6.125	0.000	0.001
ext_other	NA	-0.127	NA	0.025	NA	-0.117	NA	-5.053	NA	0.001
good	NA	0.200	NA	0.099	NA	0.043	NA	2.019	NA	0.044

District 7 | Model II Beta Coef. Comparison: SPSS vs R



Model III

Table 5: District 7 | LN_PRICE Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1805	1779	26	0.4302	0.3420	0.3320	35.4870
R	1805	1779	26	0.4317	0.3368	0.3272	34.7546
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.35%	-1.52%	-1.45%	-2.06%

Table 6: District 7 | LN_PRICE Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	0.9890	1.0970	0.9140	1.2000	0.3760	
R	0.9855	1.0974	0.9137	1.2011	0.3771	-0.4988
% diff (R vs SPSS)	-0.35%	0.04%	-0.03%	0.09%	0.28%	

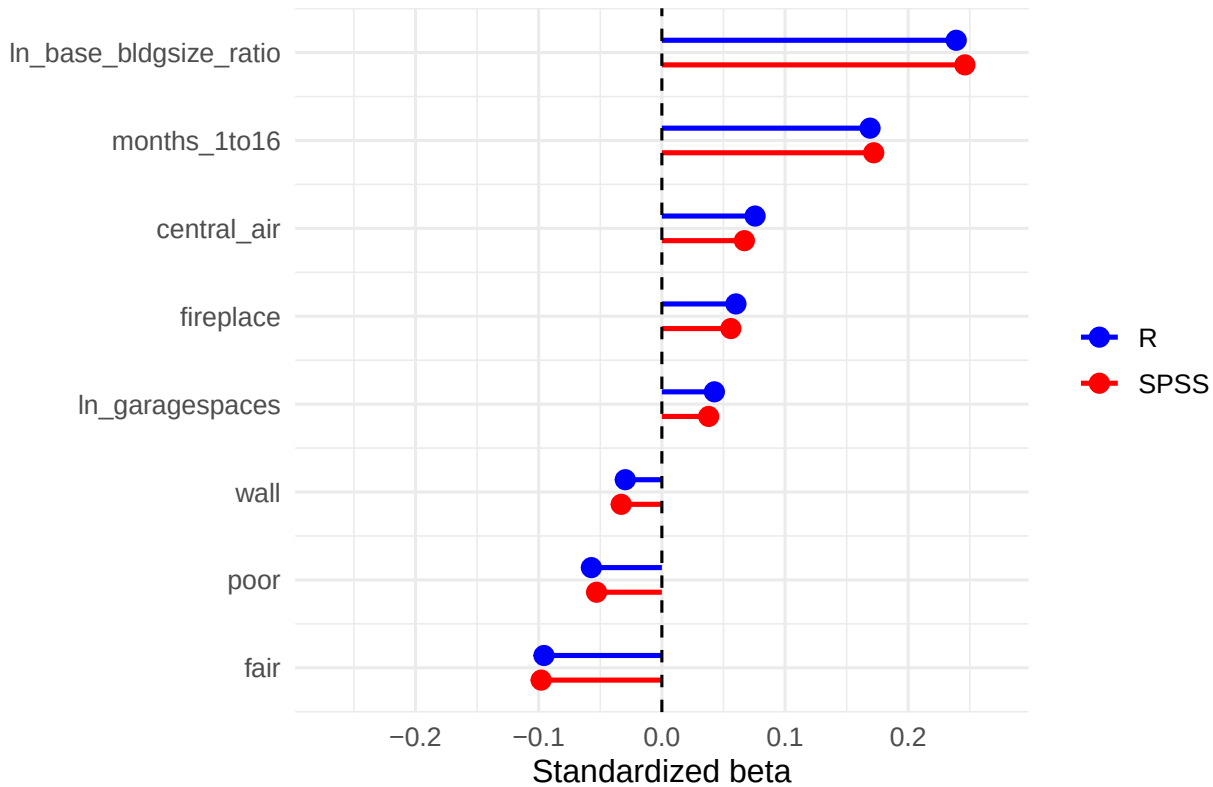
Table 7: District 7 | Model III Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	10.8640	NA	0.0391	NA	NA	NA	277.5178	NA
ecf7r701	0.5986	0.440	0.0596	0.056	0.2103	0.155	10.0508	7.793
ln_base_bldgsize_ratio	0.5234	0.540	0.0626	0.056	0.2390	0.246	8.3645	9.690
ecf7r702	0.5030	0.415	0.0503	0.047	0.2134	0.176	10.0047	8.851
ecf7r717	-0.4969	-0.614	0.0761	0.071	-0.1456	-0.180	-6.5319	-8.681
ecf7r723	0.4862	0.362	0.0772	0.073	0.1440	0.107	6.3009	4.967
ecf7r721	0.4247	0.260	0.1485	0.147	0.0568	0.035	2.8590	1.770
ecf7r716	-0.4075	-0.543	0.0858	0.082	-0.0974	-0.130	-4.7506	-6.597
ecf7r715	-0.3898	-0.490	0.1468	0.146	-0.0522	-0.066	-2.6562	-3.369
ecf7r722	-0.3716	-0.485	0.1439	0.140	-0.0524	-0.068	-2.5829	-3.459
ecf7r718	-0.2461	-0.381	0.0620	0.058	-0.0831	-0.129	-3.9697	-6.548
ecf7r710	0.2412	0.113	0.0523	0.049	0.0976	0.046	4.6142	2.325
wall	-0.2219	-0.245	0.1458	0.145	-0.0297	-0.033	-1.5215	-1.691
poor	-0.1800	-0.167	0.0620	0.062	-0.0572	-0.053	-2.9025	-2.702
ecf7r719	0.1748	NA	0.0393	NA	0.1045	NA	4.4458	NA
slab	-0.1553	NA	0.1014	NA	-0.0301	NA	-1.5315	NA
fair	-0.1292	-0.133	0.0268	0.027	-0.0957	-0.098	-4.8234	-4.992
ecf7r704	0.1179	NA	0.0376	NA	0.0715	NA	3.1355	NA
yb_pre_1929	-0.1104	NA	0.0400	NA	-0.0732	NA	-2.7602	NA
ecf7r705	0.1084	NA	0.0301	NA	0.0926	NA	3.6025	NA
central_air	0.1034	0.092	0.0272	0.027	0.0757	0.067	3.8066	3.411
hot_water	0.1025	NA	0.0559	NA	0.0433	NA	1.8321	NA
ln_garagespaces	0.0720	0.064	0.0336	0.034	0.0426	0.038	2.1431	1.911
fireplace	0.0714	0.067	0.0285	0.028	0.0601	0.056	2.5061	2.395
qbelow_avg	-0.0614	NA	0.0260	NA	-0.0571	NA	-2.3616	NA
yb_1930_to_1945	-0.0428	NA	0.0246	NA	-0.0380	NA	-1.7446	NA
months_1to16	0.0169	0.017	0.0019	0.002	0.1690	0.172	8.7009	8.849
(intercept)	NA	10.960	NA	0.030	NA	NA	NA	360.689
ln_landratio	NA	0.117	NA	0.064	NA	0.039	NA	1.834
ext_other	NA	-0.113	NA	0.024	NA	-0.105	NA	-4.743
ecf7r703	NA	-0.189	NA	0.071	NA	-0.052	NA	-2.652
ecf7r707	NA	-0.209	NA	0.109	NA	-0.037	NA	-1.923
ecf7r708	NA	-0.125	NA	0.054	NA	-0.045	NA	-2.311
ecf7r711	NA	-0.175	NA	0.047	NA	-0.076	NA	-3.700
ecf7r720	NA	-0.279	NA	0.165	NA	-0.033	NA	-1.690
ecf7r724	NA	-0.223	NA	0.107	NA	-0.041	NA	-2.083

Table 8: District 7 | Model III Coefficient Comparison: SPSS vs R

term	Sig		Tolerance		VIF	
	R	SPSS	R	SPSS	R	SPSS
(constant)	0.0000	NA	NA	NA	NA	NA
ecf7r701	0.0000	0.001	0.8513	0.941	1.1747	1.063
ln_base_bldgsize_ratio	0.0000	0.001	0.4567	0.572	2.1898	1.747
ecf7r702	0.0000	0.001	0.8194	0.934	1.2204	1.071
ecf7r717	0.0000	0.001	0.7502	0.861	1.3330	1.162
ecf7r723	0.0000	0.001	0.7132	0.795	1.4021	1.257
ecf7r721	0.0043	0.077	0.9432	0.958	1.0603	1.044
ecf7r716	0.0000	0.001	0.8876	0.957	1.1266	1.045
ecf7r715	0.0080	0.001	0.9663	0.976	1.0348	1.024
ecf7r722	0.0099	0.001	0.9057	0.948	1.1041	1.055
ecf7r718	0.0001	0.001	0.8499	0.958	1.1765	1.044
ecf7r710	0.0000	0.020	0.8330	0.960	1.2004	1.041
wall	0.1283	0.091	0.9789	0.986	1.0216	1.014
poor	0.0037	0.007	0.9597	0.961	1.0420	1.040
ecf7r719	0.0000	NA	0.6748	NA	1.4819	NA
slab	0.1258	NA	0.9636	NA	1.0378	NA
fair	0.0000	0.001	0.9472	0.953	1.0557	1.050
ecf7r704	0.0017	NA	0.7159	NA	1.3968	NA
yb_pre_1929	0.0058	NA	0.5299	NA	1.8871	NA
ecf7r705	0.0003	NA	0.5636	NA	1.7742	NA
central_air	0.0001	0.001	0.9428	0.950	1.0606	1.052
hot_water	0.0671	NA	0.6685	NA	1.4959	NA
ln_garagespaces	0.0322	0.056	0.9418	0.928	1.0618	1.077
fireplace	0.0123	0.017	0.6473	0.665	1.5449	1.503
qbelow_avg	0.0183	NA	0.6373	NA	1.5691	NA
yb_1930_to_1945	0.0812	NA	0.7867	NA	1.2711	NA
months_1to16	0.0000	0.001	0.9877	0.981	1.0124	1.019
(intercept)	NA	0.001	NA	NA	NA	NA
ln_landratio	NA	0.067	NA	0.811	NA	1.233
ext_other	NA	0.001	NA	0.762	NA	1.312
ecf7r703	NA	0.008	NA	0.976	NA	1.025
ecf7r707	NA	0.055	NA	0.985	NA	1.015
ecf7r708	NA	0.021	NA	0.965	NA	1.036
ecf7r711	NA	0.001	NA	0.878	NA	1.139
ecf7r720	NA	0.091	NA	0.977	NA	1.023
ecf7r724	NA	0.037	NA	0.962	NA	1.040

District 7 | Model III Beta Coef. Comparison: SPSS vs R



Model IV

Table 9: District 7 | LN_PRICE (Inliers) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1623	1598	25	0.3668	0.4260	0.4170	47.4560
R	1647	1621	26	0.3748	0.3936	0.3839	40.4716
% diff (R vs SPSS)	1.48%	1.44%	4.00%	2.18%	-7.61%	-7.94%	-14.72%

Table 10: District 7 | LN_PRICE (Inliers) Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0070	1.0680	0.9380	1.1390	0.3130	-0.3640
R	0.9961	1.0717	0.9358	1.1452	0.3230	-0.4079
% diff (R vs SPSS)	-1.08%	0.35%	-0.23%	0.55%	3.19%	-12.05%

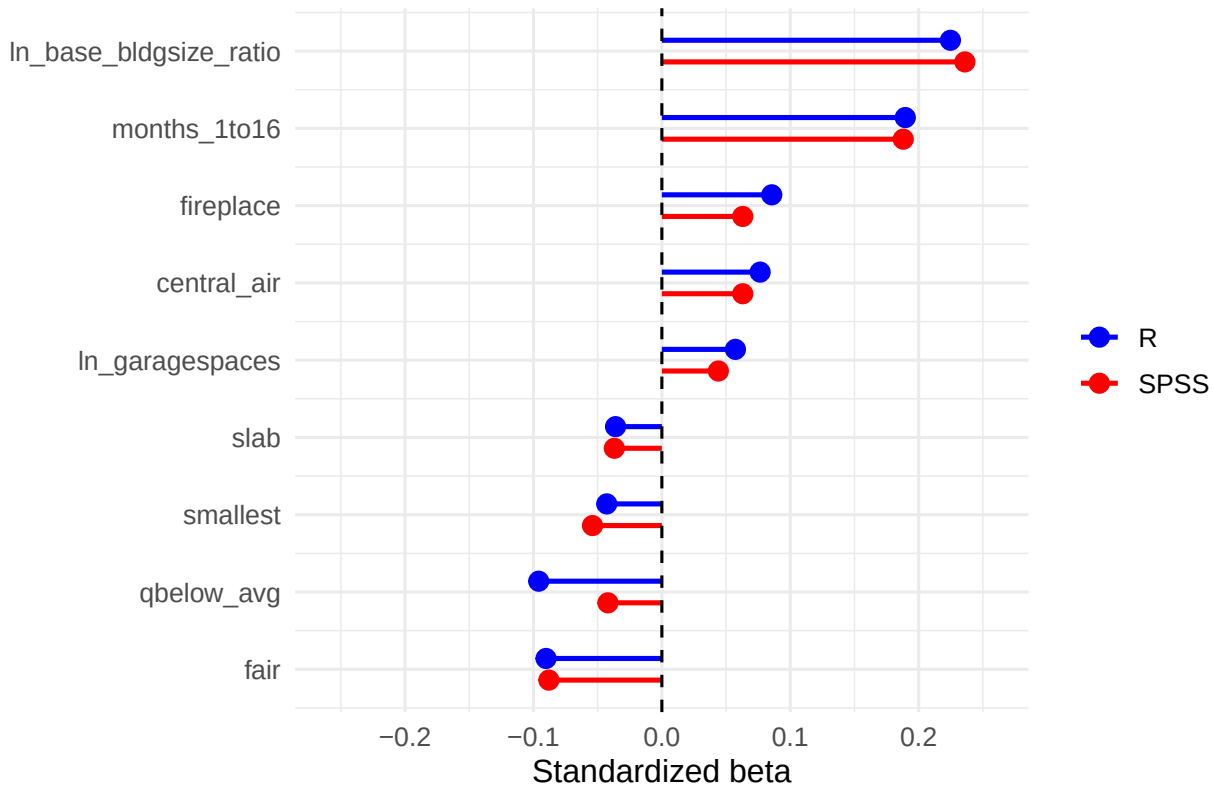
Table 11: District 7 | Model IV Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	10.8941	NA	0.0369	NA	NA	NA	295.5923	NA
ecf7r701	0.6206	0.516	0.0546	0.053	0.2396	0.198	11.3598	9.805
ecf7r717	-0.5983	-0.630	0.0860	0.076	-0.1502	-0.171	-6.9573	-8.298
ecf7r702	0.5116	0.479	0.0451	0.043	0.2473	0.230	11.3475	11.149
ln_base_bldgsize_ratio	0.4888	0.500	0.0757	0.067	0.2247	0.236	6.4585	7.430
ecf7r715	-0.4226	-0.481	0.2684	0.263	-0.0308	-0.035	-1.5746	-1.829
ecf7r723	0.3937	0.487	0.0908	0.078	0.1028	0.137	4.3333	6.281
ecf7r716	-0.3865	-0.498	0.1036	0.109	-0.0743	-0.089	-3.7313	-4.583
ecf7r718	-0.3096	-0.379	0.0627	0.058	-0.1046	-0.131	-4.9383	-6.492
ecf7r710	0.3079	0.223	0.0491	0.046	0.1336	0.099	6.2693	4.860
ecf7r720	-0.2257	NA	0.1451	NA	-0.0308	NA	-1.5550	NA
slab	-0.2114	-0.178	0.1150	0.095	-0.0361	-0.037	-1.8374	-1.866
two_baths	0.2004	NA	0.1091	NA	0.0436	NA	1.8365	NA
ecf7r719	0.1858	0.087	0.0371	0.034	0.1243	0.059	5.0029	2.526
poor	-0.1466	NA	0.0636	NA	-0.0455	NA	-2.3059	NA
fair	-0.1114	-0.110	0.0246	0.024	-0.0902	-0.088	-4.5284	-4.522
ecf7r705	0.1108	0.088	0.0279	0.025	0.1062	0.084	3.9723	3.508
ecf7r704	0.1102	NA	0.0344	NA	0.0762	NA	3.2057	NA
yb_pre_1929	-0.0979	NA	0.0367	NA	-0.0638	NA	-2.6656	NA
fireplace	0.0947	0.070	0.0268	0.027	0.0856	0.063	3.5390	2.638
qbelow_avg	-0.0943	-0.042	0.0241	0.024	-0.0959	-0.042	-3.9089	-1.767
central_air	0.0942	0.079	0.0243	0.024	0.0765	0.063	3.8817	3.277
ln_garagespaces	0.0877	0.067	0.0306	0.030	0.0573	0.044	2.8718	2.229
ecf7r711	-0.0753	-0.151	0.0453	0.043	-0.0377	-0.075	-1.6631	-3.505
smallest	-0.0619	-0.079	0.0370	0.035	-0.0429	-0.054	-1.6731	-2.244
yb_1930_to_1945	-0.0440	NA	0.0222	NA	-0.0433	NA	-1.9790	NA
months_1to16	0.0171	0.017	0.0018	0.002	0.1896	0.188	9.7401	9.842
(intercept)	NA	10.952	NA	0.032	NA	NA	NA	340.335
ln_landratio	NA	0.204	NA	0.058	NA	0.072	NA	3.515
qpoor	NA	0.305	NA	0.167	NA	0.035	NA	1.831
ext_other	NA	-0.147	NA	0.022	NA	-0.149	NA	-6.575
ecf7r703	NA	-0.150	NA	0.073	NA	-0.041	NA	-2.066
ecf7r722	NA	-0.446	NA	0.262	NA	-0.033	NA	-1.706

Table 12: District 7 | Model IV Coefficient Comparison: SPSS vs R

term	Sig		Tolerance		VIF	
	R	SPSS	R	SPSS	R	SPSS
(constant)	0.0000	NA	NA	NA	NA	NA
ecf7r701	0.0000	0.001	0.8410	0.884	1.1891	1.131
ecf7r717	0.0000	0.001	0.8030	0.848	1.2454	1.179
ecf7r702	0.0000	0.001	0.7874	0.840	1.2701	1.190
ln_base_bldgsize_ratio	0.0000	0.001	0.3089	0.356	3.2372	2.812
ecf7r715	0.1156	0.068	0.9759	0.973	1.0247	1.028
ecf7r723	0.0000	0.001	0.6650	0.759	1.5038	1.318
ecf7r716	0.0002	0.001	0.9428	0.958	1.0606	1.043
ecf7r718	0.0000	0.001	0.8344	0.885	1.1984	1.130
ecf7r710	0.0000	0.001	0.8238	0.861	1.2139	1.161
ecf7r720	0.1201	NA	0.9565	NA	1.0455	NA
slab	0.0663	0.062	0.9713	0.935	1.0295	1.069
two_baths	0.0665	NA	0.6623	NA	1.5099	NA
ecf7r719	0.0000	0.012	0.6059	0.664	1.6504	1.507
poor	0.0212	NA	0.9610	NA	1.0406	NA
fair	0.0000	0.001	0.9428	0.950	1.0607	1.052
ecf7r705	0.0001	0.001	0.5238	0.623	1.9093	1.604
ecf7r704	0.0014	NA	0.6623	NA	1.5099	NA
yb_pre_1929	0.0078	NA	0.6532	NA	1.5310	NA
fireplace	0.0004	0.008	0.6393	0.626	1.5642	1.596
qbelow_avg	0.0001	0.077	0.6214	0.632	1.6093	1.583
central_air	0.0001	0.001	0.9620	0.958	1.0395	1.044
ln_garagespaces	0.0041	0.026	0.9401	0.929	1.0638	1.077
ecf7r711	0.0965	0.001	0.7287	0.776	1.3723	1.288
smallest	0.0945	0.025	0.5688	0.610	1.7580	1.639
yb_1930_to_1945	0.0480	NA	0.7807	NA	1.2809	NA
months_1to16	0.0000	0.001	0.9874	0.987	1.0128	1.013
(intercept)	NA	0.001	NA	NA	NA	NA
ln_landratio	NA	0.001	NA	0.864	NA	1.157
qpoor	NA	0.067	NA	0.970	NA	1.031
ext_other	NA	0.001	NA	0.703	NA	1.422
ecf7r703	NA	0.039	NA	0.896	NA	1.115
ecf7r722	NA	0.088	NA	0.983	NA	1.017

District 7 | Model IV Beta Coef. Comparison: SPSS vs R



Model V

Table 13: District 7 | LN_TASP Model I Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1623	1599	24	0.3667	0.4020	0.3930	44.7940
R	1647	1622	25	0.3751	0.3676	0.3578	37.7124
% diff (R vs SPSS)	1.48%	1.44%	4.17%	2.30%	-8.56%	-8.96%	-15.81%

Table 14: District 7 | LN_TASP Model I Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0070	1.0680	0.9370	1.1400	0.3120	-0.3900
R	0.9890	1.0720	0.9281	1.1550	0.3326	-0.4647
% diff (R vs SPSS)	-1.79%	0.37%	-0.95%	1.32%	6.60%	-19.16%

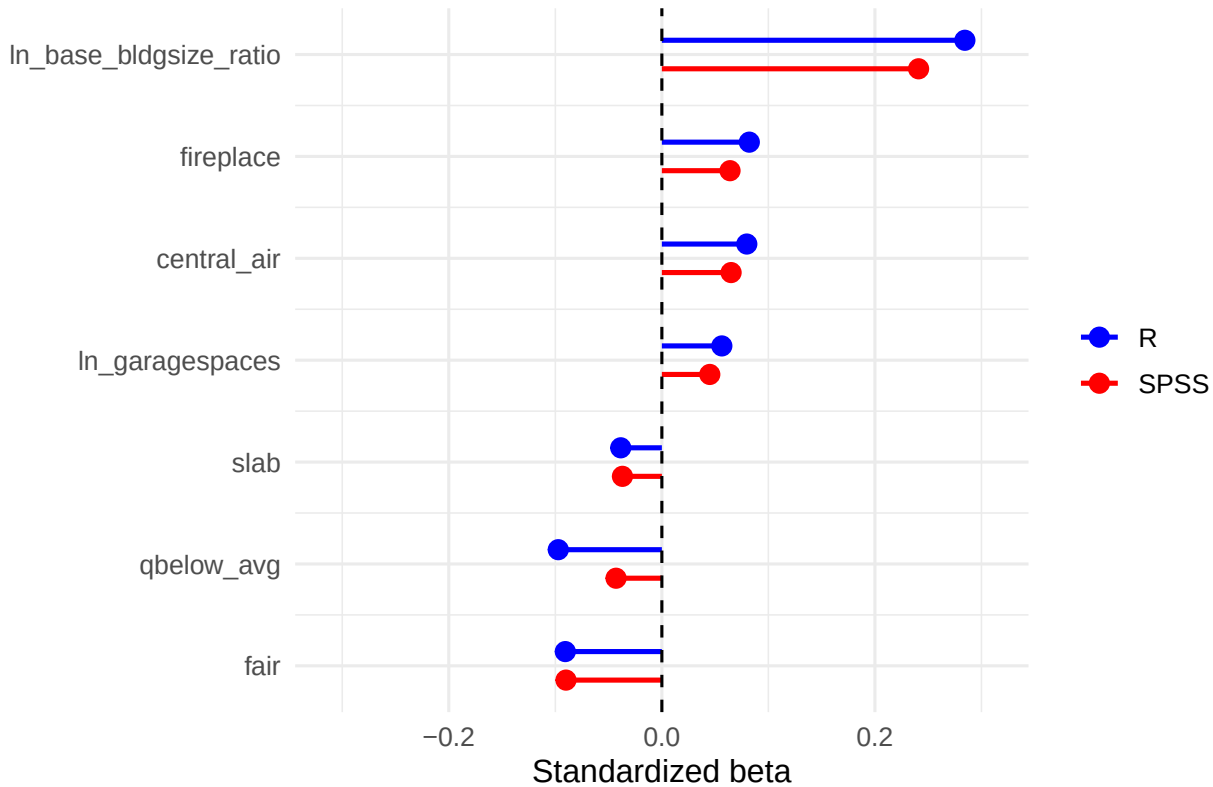
Table 15: District 7 | Model V Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.1499	NA	0.0313	NA	NA	NA	356.2988	NA
ecf7r701	0.6256	0.516	0.0546	0.053	0.2464	0.202	11.4527	9.815
ecf7r717	-0.6069	-0.630	0.0861	0.076	-0.1554	-0.174	-7.0487	-8.301
ln_base_bldgsize_ratio	0.6064	0.500	0.0643	0.067	0.2845	0.241	9.4249	7.442
ecf7r702	0.5079	0.479	0.0451	0.043	0.2505	0.235	11.2673	11.153
ecf7r715	-0.4261	-0.481	0.2679	0.263	-0.0317	-0.036	-1.5907	-1.828
ecf7r716	-0.3815	-0.498	0.1037	0.109	-0.0748	-0.091	-3.6781	-4.585
ecf7r723	0.3754	0.487	0.0905	0.078	0.1000	0.140	4.1467	6.285
ecf7r718	-0.3030	-0.379	0.0628	0.058	-0.1044	-0.134	-4.8228	-6.493
ecf7r710	0.3017	0.223	0.0490	0.046	0.1336	0.101	6.1521	4.862
ecf7r720	-0.2375	NA	0.1450	NA	-0.0330	NA	-1.6376	NA
slab	-0.2217	-0.178	0.1151	0.095	-0.0386	-0.037	-1.9262	-1.866
ecf7r719	0.1776	0.087	0.0370	0.034	0.1213	0.060	4.8072	2.527
two_baths	0.1626	NA	0.1074	NA	0.0361	NA	1.5150	NA
poor	-0.1445	NA	0.0636	NA	-0.0457	NA	-2.2714	NA
ecf7r704	0.1120	NA	0.0344	NA	0.0790	NA	3.2577	NA
fair	-0.1098	-0.110	0.0246	0.024	-0.0906	-0.090	-4.4551	-4.523
ecf7r705	0.1090	0.088	0.0278	0.025	0.1066	0.086	3.9171	3.509
yb_pre_1929	-0.0977	NA	0.0368	NA	-0.0650	NA	-2.6532	NA
central_air	0.0962	0.079	0.0243	0.024	0.0797	0.065	3.9637	3.279
qbelow_avg	-0.0937	-0.042	0.0241	0.024	-0.0972	-0.043	-3.8945	-1.767
fireplace	0.0890	0.070	0.0266	0.027	0.0821	0.064	3.3418	2.637
ln_garagespaces	0.0845	0.067	0.0306	0.030	0.0563	0.045	2.7646	2.229
ecf7r711	-0.0754	-0.151	0.0454	0.043	-0.0384	-0.077	-1.6607	-3.505
single_family2story	-0.0645	NA	0.0350	NA	-0.0432	NA	-1.8423	NA
yb_1930_to_1945	-0.0432	NA	0.0223	NA	-0.0434	NA	-1.9370	NA
(intercept)	NA	11.226	NA	0.026	NA	NA	NA	438.363
ln_landratio	NA	0.204	NA	0.058	NA	0.073	NA	3.515
smallest	NA	-0.078	NA	0.035	NA	-0.056	NA	-2.243
qpoor	NA	0.306	NA	0.167	NA	0.036	NA	1.832
ext_other	NA	-0.147	NA	0.022	NA	-0.152	NA	-6.583
ecf7r703	NA	-0.150	NA	0.073	NA	-0.042	NA	-2.069
ecf7r722	NA	-0.446	NA	0.262	NA	-0.033	NA	-1.705

Table 16: District 7 | Model V Coefficient Comparison: SPSS vs R

term	Sig		Tolerance		VIF	
	R	SPSS	R	SPSS	R	SPSS
(constant)	0.0000	NA	NA	NA	NA	NA
ecf7r701	0.0000	0.001	0.8424	0.885	1.1871	1.130
ecf7r717	0.0000	0.001	0.8022	0.849	1.2466	1.178
ln_base_bldgsize_ratio	0.0000	0.001	0.4280	0.356	2.3364	2.807
ecf7r702	0.0000	0.001	0.7887	0.840	1.2679	1.190
ecf7r715	0.1119	0.068	0.9813	0.974	1.0191	1.027
ecf7r716	0.0002	0.001	0.9417	0.958	1.0619	1.043
ecf7r723	0.0000	0.001	0.6709	0.759	1.4906	1.317
ecf7r718	0.0000	0.001	0.8321	0.885	1.2018	1.130
ecf7r710	0.0000	0.001	0.8274	0.863	1.2087	1.159
ecf7r720	0.1017	NA	0.9595	NA	1.0422	NA
slab	0.0543	0.062	0.9715	0.935	1.0293	1.069
ecf7r719	0.0000	0.012	0.6128	0.664	1.6319	1.506
two_baths	0.1300	NA	0.6856	NA	1.4587	NA
poor	0.0233	NA	0.9613	NA	1.0403	NA
ecf7r704	0.0011	NA	0.6628	NA	1.5088	NA
fair	0.0000	0.001	0.9420	0.950	1.0616	1.052
ecf7r705	0.0001	0.001	0.5269	0.623	1.8979	1.604
yb_pre_1929	0.0081	NA	0.6502	NA	1.5379	NA
central_air	0.0001	0.001	0.9635	0.958	1.0379	1.044
qbelow_avg	0.0001	0.077	0.6254	0.632	1.5990	1.583
fireplace	0.0009	0.008	0.6459	0.627	1.5483	1.594
ln_garagespaces	0.0058	0.026	0.9399	0.929	1.0640	1.076
ecf7r711	0.0970	0.001	0.7273	0.777	1.3750	1.288
single_family2story	0.0656	NA	0.7090	NA	1.4104	NA
yb_1930_to_1945	0.0529	NA	0.7769	NA	1.2872	NA
(intercept)	NA	0.001	NA	NA	NA	NA
ln_landratio	NA	0.001	NA	0.865	NA	1.157
smallest	NA	0.025	NA	0.611	NA	1.638
qpoor	NA	0.067	NA	0.970	NA	1.031
ext_other	NA	0.001	NA	0.704	NA	1.421
ecf7r703	NA	0.039	NA	0.897	NA	1.115
ecf7r722	NA	0.088	NA	0.984	NA	1.016

District 7 | Model V Beta Coef. Comparison: SPSS vs R



Model VI

Table 17: District 7 | LN_TASP Model II Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	2483	2452	31	0.4038	0.3920	0.3850	51.0820
R	2665	2634	31	0.4434	0.3346	0.3268	42.7269
% diff (R vs SPSS)	7.33%	7.42%	0.00%	9.81%	-14.64%	-15.12%	-16.36%

Table 18: District 7 | LN_TASP Model II Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	0.9970	1.0840	0.9280	1.1680	0.3480	-0.4410
R	0.9508	1.0708	0.8799	1.2169	0.3956	-0.5375
% diff (R vs SPSS)	-4.63%	-1.22%	-5.18%	4.19%	13.67%	-21.89%

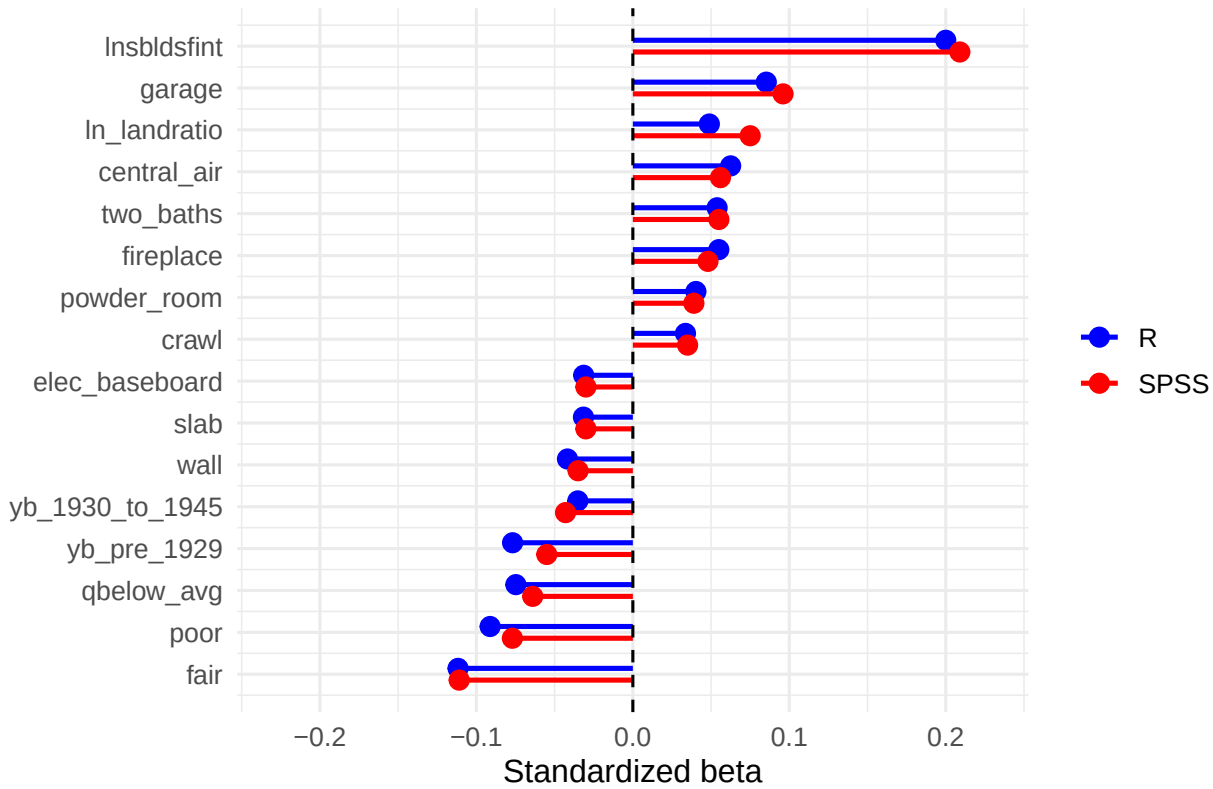
Table 19: District 7 | Model VI Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	9.4631	NA	0.1956	NA	NA	NA	48.3904	NA
ecf7r701	0.5807	0.498	0.0555	0.052	0.1789	0.159	10.4701	9.551
elec_baseboard	-0.5062	-0.439	0.2592	0.237	-0.0314	-0.030	-1.9533	-1.850
ecf7r702	0.4782	0.461	0.0463	0.042	0.1813	0.187	10.3174	11.029
ecf7r716	-0.4626	-0.529	0.0600	0.063	-0.1330	-0.143	-7.7039	-8.440
ecf7r723	0.4106	0.367	0.0745	0.075	0.1099	0.092	5.5117	4.926
ecf7r717	-0.3894	-0.461	0.0600	0.061	-0.1199	-0.138	-6.4892	-7.557
ecf7r721	0.3839	0.405	0.1391	0.238	0.0456	0.027	2.7594	1.704
ecf7r720	-0.3172	-0.399	0.1211	0.130	-0.0424	-0.049	-2.6198	-3.063
ecf7r724	-0.2838	-0.370	0.0749	0.076	-0.0646	-0.082	-3.7913	-4.845
poor	-0.2821	-0.240	0.0501	0.050	-0.0912	-0.077	-5.6368	-4.771
lnsbldsfint	0.2536	0.259	0.0303	0.029	0.2000	0.209	8.3585	8.969
wall	-0.2297	-0.203	0.0959	0.099	-0.0418	-0.035	-2.3952	-2.052
ecf7r710	0.2266	0.211	0.0470	0.044	0.0831	0.081	4.8250	4.841
two_baths	0.2259	0.249	0.0832	0.084	0.0539	0.055	2.7150	2.958
ecf7r718	-0.2116	-0.281	0.0450	0.043	-0.0831	-0.114	-4.7020	-6.505
ecf7r715	-0.1801	NA	0.0954	NA	-0.0308	NA	-1.8879	NA
ecf7r719	0.1707	0.095	0.0323	0.030	0.0993	0.059	5.2842	3.157
slab	-0.1478	-0.137	0.0783	0.074	-0.0316	-0.030	-1.8866	-1.851
fair	-0.1455	-0.138	0.0214	0.020	-0.1117	-0.111	-6.7992	-6.795
ln_landratio	0.1453	0.219	0.0546	0.051	0.0489	0.075	2.6637	4.267
crawl	0.1353	0.140	0.0706	0.067	0.0336	0.035	1.9161	2.070
yb_post_1961	0.1139	NA	0.0629	NA	0.0302	NA	1.8100	NA
yb_pre_1929	-0.1096	-0.077	0.0309	0.030	-0.0769	-0.055	-3.5486	-2.605
garage	0.0986	0.104	0.0192	0.018	0.0853	0.096	5.1328	5.748
central_air	0.0943	0.081	0.0247	0.023	0.0625	0.056	3.8193	3.466
ecf7r704	0.0940	NA	0.0333	NA	0.0540	NA	2.8192	NA
ecf7r705	0.0874	0.072	0.0256	0.023	0.0709	0.062	3.4110	3.186
powder_room	0.0861	0.084	0.0410	0.040	0.0403	0.039	2.1003	2.088
qbelow_avg	-0.0826	-0.068	0.0219	0.021	-0.0748	-0.064	-3.7617	-3.220
fireplace	0.0672	0.058	0.0239	0.023	0.0549	0.048	2.8100	2.477
yb_1930_to_1945	-0.0406	-0.047	0.0212	0.020	-0.0352	-0.043	-1.9153	-2.386
(intercept)	NA	9.540	NA	0.186	NA	NA	NA	51.212
ext_other	NA	-0.131	NA	0.020	NA	-0.126	NA	-6.562
ecf7r703	NA	-0.138	NA	0.066	NA	-0.034	NA	-2.088
ecf7r711	NA	-0.123	NA	0.039	NA	-0.057	NA	-3.189

Table 20: District 7 | Model VI Coefficient Comparison: SPSS vs R

term	Sig		Tolerance		VIF	
	R	SPSS	R	SPSS	R	SPSS
(constant)	0.0000	NA	NA	NA	NA	NA
ecf7r701	0.0000	0.001	0.8655	0.892	1.1554	1.121
elec_baseboard	0.0509	0.064	0.9766	0.965	1.0239	1.036
ecf7r702	0.0000	0.001	0.8181	0.860	1.2223	1.163
ecf7r716	0.0000	0.001	0.8471	0.862	1.1805	1.159
ecf7r723	0.0000	0.001	0.6351	0.711	1.5747	1.406
ecf7r717	0.0000	0.001	0.7395	0.748	1.3522	1.336
ecf7r721	0.0058	0.089	0.9271	0.962	1.0786	1.040
ecf7r720	0.0088	0.002	0.9627	0.966	1.0388	1.035
ecf7r724	0.0002	0.001	0.8688	0.861	1.1510	1.162
poor	0.0000	0.001	0.9646	0.951	1.0367	1.051
lnsbldsfint	0.0000	0.001	0.4413	0.458	2.2661	2.184
wall	0.0167	0.040	0.8305	0.837	1.2041	1.195
ecf7r710	0.0000	0.001	0.8525	0.885	1.1731	1.130
two_baths	0.0067	0.003	0.6420	0.709	1.5576	1.410
ecf7r718	0.0000	0.001	0.8086	0.809	1.2367	1.236
ecf7r715	0.0591	NA	0.9470	NA	1.0560	NA
ecf7r719	0.0000	0.002	0.7159	0.702	1.3969	1.425
slab	0.0593	0.064	0.9023	0.913	1.1083	1.095
fair	0.0000	0.001	0.9359	0.925	1.0685	1.081
ln_landratio	0.0078	0.001	0.7498	0.792	1.3337	1.262
crawl	0.0555	0.039	0.8196	0.847	1.2202	1.180
yb_post_1961	0.0704	NA	0.9059	NA	1.1039	NA
yb_pre_1929	0.0004	0.009	0.5385	0.558	1.8570	1.792
garage	0.0000	0.001	0.9148	0.887	1.0932	1.128
central_air	0.0001	0.001	0.9428	0.942	1.0606	1.062
ecf7r704	0.0048	NA	0.6889	NA	1.4516	NA
ecf7r705	0.0007	0.001	0.5844	0.645	1.7111	1.549
powder_room	0.0358	0.037	0.6860	0.716	1.4578	1.397
qbelow_avg	0.0002	0.001	0.6396	0.628	1.5636	1.592
fireplace	0.0050	0.013	0.6610	0.649	1.5129	1.542
yb_1930_to_1945	0.0556	0.017	0.7493	0.766	1.3347	1.305
(intercept)	NA	0.001	NA	NA	NA	NA
ext_other	NA	0.001	NA	0.676	NA	1.479
ecf7r703	NA	0.037	NA	0.927	NA	1.079
ecf7r711	NA	0.001	NA	0.779	NA	1.283

District 7 | Model VI Beta Coef. Comparison: SPSS vs R

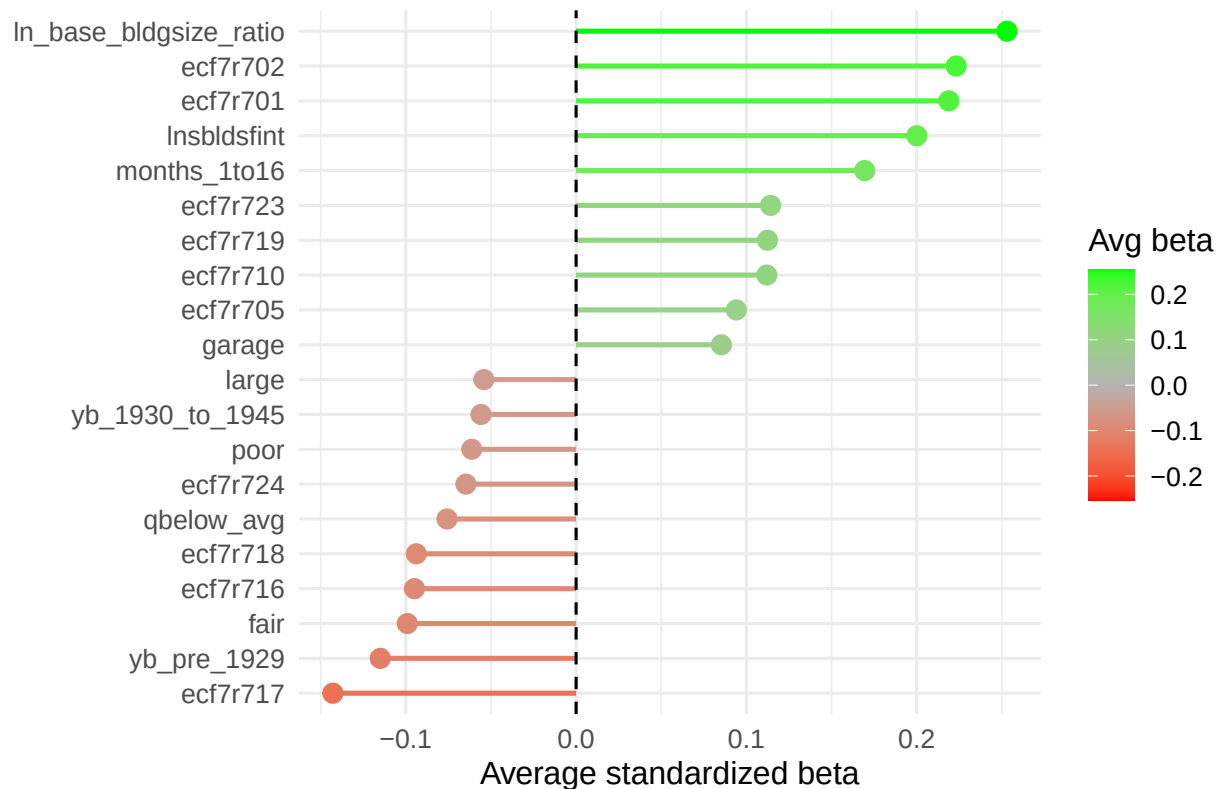


Model Comparison

Table 21: Top and Bottom Standardized Beta Predictors across Models

Predictor	n	Models	Avg b	Avg SE	Avg beta	Avg t	Avg sig	Min beta	Max beta
In_base_bldgsize_ratio	4	M2, M3, M4, M5	0.549	0.067	0.253	8.313	0.000	0.225	0.284
ecf7r702	4	M3, M4, M5, M6	0.500	0.047	0.223	10.734	0.000	0.181	0.251
ecf7r701	4	M3, M4, M5, M6	0.606	0.056	0.219	10.833	0.000	0.179	0.246
Insbldsfint	1	M6	0.254	0.030	0.200	8.359	0.000	0.200	0.200
months_1to16	2	M1, M2	0.015	0.002	0.159	6.914	0.000	0.149	0.170
central_air	2	M1, M2	0.132	0.030	0.102	4.382	0.000	0.097	0.106
In_garagespaces	1	M1	0.140	0.039	0.092	3.627	0.000	0.092	0.092
yb_1930_to_1945	2	M1, M2	-0.094	0.027	-0.088	-3.485	0.001	-0.103	-0.072
qbelow_avg	3	M1, M4, M5	-0.089	0.026	-0.091	-3.500	0.002	-0.097	-0.081
ecf7r718	2	M4, M5	-0.306	0.063	-0.104	-4.881	0.000	-0.105	-0.104
fair	3	M2, M3, M6	-0.140	0.026	-0.105	-5.555	0.000	-0.112	-0.096
ecf7r716	2	M3, M6	-0.435	0.073	-0.115	-6.227	0.000	-0.133	-0.097
ecf7r717	4	M3, M4, M5, M6	-0.523	0.077	-0.143	-6.757	0.000	-0.155	-0.120
yb_pre_1929	2	M1, M2	-0.306	0.044	-0.205	-6.979	0.000	-0.216	-0.195

District 7 | Average Standardized Beta across Models



Prediction Testing

```
## [1] "22024510."
```

```
## [1] 90675.86
```

Table 22: Deviation Summary: LN_SPPSF Model

Metric	Value
Average sale price deviation	-5849.30
Max deviation	157055.25
Min deviation	-866257.46
Standard deviation	42342.25

```
## [1] "22024510."
```

```
## [1] 87108.02
```

Table 23: Deviation Summary: LN_PRICE - No ECFs Model

Metric	Value
Average sale price deviation	-8034.94
Max deviation	145921.10
Min deviation	-873117.62
Standard deviation	41938.64

```
## [1] "22024510."
```

[1] 95795.85

Table 24: Deviation Summary: LN_PRICE Model

Metric	Value
Average sale price deviation	-6792.60
Max deviation	90781.78
Min deviation	-869263.59
Standard deviation	39152.96

[1] "22024510."

[1] 101416.3

Table 25: Deviation Summary: LN_PRICE - Outliers Model

Metric	Value
Average sale price deviation	-4990.95
Max deviation	80079.29
Min deviation	-104414.46
Standard deviation	28890.54

[1] "22024510."

[1] 108961.7

Table 26: Deviation Summary: LN_TASP Model I

Metric	Value
Average sale price deviation	-330.72
Max deviation	95861.31
Min deviation	-103903.55
Standard deviation	29321.97

[1] "22024510."

[1] 101334.5

Table 27: Deviation Summary: LN_TASP Model II

Metric	Value
Average sale price deviation	-1550.10
Max deviation	86854.77
Min deviation	-127890.17
Standard deviation	29620.32

Table 28: Deviation Summary: Time-Adjusted Model I

Metric	Value
Number of observations	1648.0
Average sale price deviation	80321800.3
Median sale price deviation	67949334.5
Mean absolute deviation	80321800.3
Median absolute deviation	67949334.5
RMSE	94075142.7
Max deviation	517726411.3
Min deviation	22143051.0
Standard deviation	48989766.0
Average percentage deviation	113104.1
Median percentage deviation	101568.4
Average absolute percentage deviation	113104.1
Median absolute percentage deviation	101568.4

Table 29: Deviation Summary: Time-Adjusted Model II

Metric	Value
Number of observations	1648.00
Average sale price deviation	78690907.16
Median sale price deviation	67785117.33
Mean absolute deviation	78690907.16
Median absolute deviation	67785117.33
RMSE	90981693.77
Max deviation	509295264.38
Min deviation	20981961.15
Standard deviation	45680147.97
Average percentage deviation	111665.86
Median percentage deviation	99924.63
Average absolute percentage deviation	111665.86
Median absolute percentage deviation	99924.63